EXTRAI: correcting and creating meta-analyses with local language models — a pre-registered five-phase benchmark with a perturbation reading proof and a self-correcting answer key

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Abstract

Language models are entering systematic-review workflows, but most evaluations use cloud models, cannot separate reading from training-data recall, and treat the human answer key as infallible. EXTRAI is a pre-registered, five-phase benchmark of five quantized open-weight models (8–14.8B parameters, one consumer mini-PC, integrated GPU, no cloud calls) as meta-analysis workers over two published meta-analyses and their 21 randomized trials, under three controls: a *perturbation reading proof* (load-bearing numbers displaced under sealed maps, so reciting a published value is detectable); a *two-layer, source-verified answer key* with a quotation-backed adjudication rite that puts the human key itself on trial — accumulating seventeen source-confirmed errata against one published anchor, and correcting its own cells twice; and an *instruction-language comparison* against the study's archived development pilot, which ran identical materials under Portuguese instruments. The five phases ascend from reading to deployment. Reading: 140 perturbed sheets, zero attributable recitations, replicate stability 77–99%. Arithmetic: 1 of 57 unaided confidence intervals exact, rising to 82% through a trivial text-protocol calculator; pooled calls stayed beyond the whole size class. Creation: *all five* models' sheets reproduced the published dichotomous pools (morbidity 0.771–0.779 vs. 0.778), while the continuous diamond separated them — exactly one model, `gemma4:12b`, landed beside the published -0.24 $[-0.32, -0.16]$, at -0.27; the rest landed -0.47 to -0.80 for named reading failures, never arithmetic. Orchestration: under a warn-only ten-net detection harness, warnings were resolved by confirm-or-correct and the model's own pooling call closed digit-consistent with its executed results. Deployment, on clean texts with no answer key in any loop: the model sided with the primary sources at all seven source-confirmed errata of the published meta-analysis; its one failed tolerance decomposed to a single mislabeled-dispersion cell seeded by the source's own two-layer notation; and a three-configuration comparison showed conversational and sheet-level detection nets are complements, not substitutes. Every phase's result sat within replicate-level variance of the pilot — the flagship diamond identical to the hundredth (-0.27 in both) — so none of the conclusions is an artifact of instruction language. All protocols, amendments, seals, sheets, graders, adjudications and raw outputs are public and archived.

1 Introduction

Systematic reviews and meta-analyses are the reference standard of clinical evidence synthesis, and they are labor-bound: extracting structured data from primary trials, checking it, and computing pooled effects consume the majority of a review team's hours. Large language models are entering exactly this workflow. Feasibility studies report promising but uneven accuracy

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for trial data extraction [Schmidt et al., 2024]; assisted extraction has proved noninferior to human-only extraction inside a live review [Lee et al., 2026]; screening assistance is approaching routine use [Xie et al., 2026]; and end-to-end platforms now report complete systematic reviews produced in hours [Bakker et al., 2026], with fully automated drafts rated above a published human-authored review by blinded specialists [McLaughlin et al., 2026]. The major evidence-synthesis organizations have jointly conditioned such use on demonstrated rigor, human oversight and transparent reporting [Flemyng et al., 2025].

Three gaps run through this literature. First, most evaluations use commercial cloud models, while clinical text pushes in the opposite direction: locally run open-weight models keep sensitive material on-premises and have shown extraction performance approaching expert annotation [Wiest et al., 2024, Spaanderman et al., 2025]. Second, benchmarks that score agreement with a published review cannot distinguish a model that *read* the primary trial from one that *remembers* the review — both the anchor and its primaries are plausibly in training corpora, and contamination inflates apparent competence [Xu et al., 2024]. Third, the published review itself is usually treated as an infallible answer key, although hand-made evidence tables carry their own errors; a benchmark that cannot notice when the published value is the one that is wrong will misgrade its best readers.

The EXTRAI benchmark was built around those three gaps: quantized open-weight models on a consumer machine with no cloud calls; a perturbation reading proof, in which load-bearing numbers are displaced in every source text under sealed maps so that recitation of published values is detectable; and a two-layer, source-verified answer key with a quotation-backed adjudication rite that puts the human key itself on trial — a verification process that, during instrument development, accumulated seventeen source-confirmed errata against one published anchor while also logging the adjudicators' own errors in the same public file [Barra, 2026]. Instrument development and validation were piloted in Brazilian Portuguese, the research team's working language; the definitive campaign reported here runs entirely under English instruments — the language the evaluated models were predominantly trained on — with the archived pilot record retained so that instruction-language effects could be bounded by direct per-phase comparison rather than assumed away.

In this study, we conducted a single pre-registered, five-phase benchmark of local language models as meta-analysis workers — reading, arithmetic, creation of two published meta-analyses, orchestration under a warn-only detection harness, and deployment on clean texts with no answer key in any loop — comparing five integrated-GPU models under contamination-controlled, source-verified grading, and measuring at each phase both the models' performance against the published values and the robustness of every result to the instruction-language change from the development pilot.

2 Methods

2.1 Study design and pre-registration

This was a pre-registered, five-phase benchmark. Its corpora, sealed perturbations, answer keys, deterministic engines, graders and detection harness were built and validated during a development pilot conducted under Portuguese-language instruments (the team's working language), whose complete record — protocols, runs, gradings and adjudications — is archived alongside this study's [Barra, 2026]. The definitive campaign reported here ran entirely under English instruments on those frozen materials, so that the pilot doubles as an instruction-language comparison arm: for each phase, the only variable separating the campaign from the pilot is the language of the instruments. The campaign protocol — phase order, model cast, hypotheses and compute budget — was registered and committed to the public repository before any run; the two extensions executed within the deployment phase were likewise registered as dated amendments before their runs. All prompts, seals, keys, harness code, graders, raw model outputs, run logs and adjudication records are public.

The campaign comprised five phases in an ascending line of investigation: P1 *Read* (multi-model structured extraction under the perturbation reading proof), P2 *Calculate* (meta-analytic arithmetic unaided versus through a text-protocol calculator), P3 *Create* (deterministic engines assemble both meta-analyses from each model's sheets, with an erratum-aware comparison against the published values), P4 *Orchestrate* (the best-performing reader executes its own calculations under a warn-only detection harness), and P5 *Deploy* (clean, unperturbed texts with no answer key in any loop, including a three-configuration comparison of error-detection architectures).

2.2 Anchor meta-analyses and corpora

Two recently published meta-analyses served as anchors. Anchor 1 addresses goal-directed fluid therapy (GDFT) versus conventional fluid therapy in major surgery [Ashraf et al., 2026]; its 14 primary randomized controlled trials constitute the first corpus (8 open-access, obtained from PubMed Central; 6 paywalled, legally obtained and never redistributed — scripts in the repository rebuild that stratum from a reader's own copies). Anchor 2 addresses low-carbohydrate diets and glycemic control (HbA1c) in type 2 diabetes [Khraise, 2026]; its 7 primary trials constitute the second corpus. Anchor 1 supplies dichotomous outcome families (overall morbidity, mortality, postoperative ileus) and two continuous ones; Anchor 2 supplies a single continuous outcome (change in HbA1c, %), whose published random-effects pooled estimate, -0.24 $[-0.32, -0.16]$, is the study's external reference value.

2.3 The perturbation reading proof

Before any model reads a trial, a sealed, deterministic operator displaces load-bearing numbers in its text. A model that returns a displaced value read the text in front of it; one that returns the original published value is reciting training data [Xu et al., 2024] — without this control, agreement with a published meta-analysis cannot be distinguished from memory of it. Models operate exclusively in the perturbed world; the original↔perturbed maps are held by the graders alone and applied only after extraction, as a pure table lookup (the "unperturbation lens"). The sealed maps were produced once, in the pilot, and reused unchanged; their SHA-256 fingerprints were printed into each phase's run log. Phase P5 deliberately runs on clean texts and therefore makes no reading-proof claim (Section 2.10).

2.4 Answer keys and the adjudication rite

Each corpus is graded against a two-layer, source-verified answer key built during the development pilot: layer 1 holds the anchor's published cells; layer 2 holds each cell verified against the primary's full text, with a literal supporting quotation and a per-cell verdict. The key itself is on trial by design — source verification of Anchor 1 accumulated a public errata file of 17 source-confirmed entries against the published meta-analysis (arms swapped in a characteristics table; reported values published as "not stated"; a unit conversion contradicting its own source; an outcome table whose counts belong to a different outcome; a systematic analyzed-as-randomized pattern in the sample sizes; among others), and the campaign graded against the key as corrected through the most recent adjudications. Mechanical grading is backed by an adjudication rite: no cell changes category without the source quotation that decides it, and the adjudicators' own errors are published as errata in the same file as everyone else's.

2.5 Instruments and the instruction-language comparison

The instruments were developed and validated in Brazilian Portuguese during the pilot, then translated into English for the definitive campaign — task framing, rules, JSON field names, calculator function names, markers and placeholders — following a published correspondence table (for example, `n_randomizados_gdft`→`n_randomized_gdft`; `ic95_md`→`ci95_md`; `RESULTADO`→`RESULT`). The English set was frozen into the campaign's record at registration; the Portuguese originals remain untouched in the archived pilot. Because corpora, seals, keys, engines and graders are identical between the campaign and the pilot, any per-phase difference between the two estimates the effect of instruction language alone; the pre-registered acceptance criterion was that these differences stay within replicate-level variance.

2.6 Models and hardware

Five quantized open-weight models sized for an integrated GPU formed the comparison cast: `gemma4:12b`, `qwen3:14b`, `llama3.1:8b`, `qwen3.5:9b` and `deepseek-r1:14b`, each pinned by its Ollama build digest in the repository's model manifest. All phases ran on one consumer mini-PC (AMD Ryzen 7, Radeon 780M integrated GPU, 32 GB RAM, no discrete GPU), served by Ollama with a 16,384-token context, reasoning channels disabled, and a uniform 4,000-token output allowance for extraction tasks; no cloud API was called anywhere in any evaluated pipeline. One model was resident at a time, with an explicit memory unload between models. Phase P4 used only `gemma4:12b`, the best-performing reader of both records.

2.7 Phase P1 — structured extraction

Each of the five models extracted all 14 perturbed Anchor-1 primaries from zero, twice (two independent replicates), under the frozen English extraction sheet (30 fields per trial, each an object `{"value", "where"}`). Grading used the eligible cell set of the two-layer key — cells with a usable, digit-bearing source-verified value; 124 cells per model — comparing the seal-reversed first-parseable sheet against the key's source layer with a magnitude-based numeric comparator, declared an approximation of the full ruler (it cannot see summarization equivalence or population-layer choices); all non-matching cells were listed for adjudication rather than auto-verdicted. Replicate stability was the same comparator applied between the two raw replicates. Recitation candidates were detected mechanically — a cell whose raw text contains a seal pair's original value while the displaced image the model actually read is absent — and each candidate was adjudicated against the perturbed source text before any verdict.

2.8 Phase P2 — meta-analytic arithmetic

Each model computed, over its own P1 sheets, per-study risk ratios and mean differences with 95% confidence intervals, and pooled estimates, in two arms: arm A unaided (no tool), and arm B through a plain-text calculator protocol in which the model writes lines of the form `CALC: function(arguments)`, the harness executes the validated function and returns `RESULT:` into the context (up to 5 rounds and 20 calls), and the model then answers with a final JSON. Three prompt families (risk ratios, mean differences, pools) × two arms × two replicates were run per model. Grading recomputed the ground truth with the same validated engine from the same input block the model saw, under the pilot's frozen parsing rules (event counts read from forms such as "19 (32.8%)"; percent-only cells converted against the group size; means accepted only as mean ± SD, so that median (IQR) cells have no computable truth and *not computable* is the correct answer there). Each reported quantity was labelled exact, right-direction, wrong, correctly-refused, wrongly-refused, or unsupported; intervals and pooled estimates were graded separately.

2.9 Phase P3 — deterministic creation of both meta-analyses

Everything downstream of the sheets was executed by pre-registered, frozen code; no model touched a number after extraction. For Anchor 1, the engines parsed each model's P1 sheets, selected event/denominator routes per outcome, computed per-study risk ratios and pooled them under both Mantel–Haenszel and DerSimonian–Laird [DerSimonian and Laird, 1986], applied the sealed reversal, and classified every difference from the published tables into the frozen five-category, erratum-aware comparison developed in the pilot (*reproduces / differs by a documented anchor erratum / model route / model error / source unavailable*); the pooled comparison is made under DerSimonian–Laird because the published pooled numbers instantiate that method under a Mantel–Haenszel caption (anchor erratum #15). For Anchor 2, each model freshly extracted the 7 perturbed primaries (two replicates) under the English continuous-outcome sheet; a deterministic route selector then built each trial's sextet (mean, SD, n per arm) by the frozen rules — declared change used directly; standard errors and confidence intervals converted to SDs; baseline-and-final pairs combined under the Cochrane $r = 0.5$ rule [Higgins et al., 2023] — and the DerSimonian–Laird pool was computed twice: in the perturbed world, and through the graders' sealed lens for comparison with the published -0.24 $[-0.32, -0.16]$. Comparison tolerances were pre-registered at ± 0.01 for risk ratios and interval bounds and ± 0.1 for mean differences.

2.10 Phase P4 — orchestration under a warn-only detection harness

The benchmark's ten-net detection harness, developed and frozen in the pilot — its doctrine being that nets *detect and warn, never substitute a value* — was ported to English with its net logic unchanged: schema-constrained typed calls (`{"function", "arguments", "source", "derivation"}`), an arity check with the full function signature, a sign-echo net, an arm-agnostic declared-source check, a declared-derivation check, an argument type system with per-class value impossibilities, a confidence-interval bound-order check, a coherence check tying each interval call to the study's own last executed mean difference, a negative-SD result warning, a closing-versus-executed check, per-call warning budgets with confirm-by-repeat, and an escalation flag (`requires_human_review`) on insistence. `gemma4:12b` then ran one formal pipeline over its own Anchor-2 sheets: per-study typed calls, a model-emitted DerSimonian–Laird pooling call over its own sextets, a product-layer coherence check on the reported intervals (flagged, never endorsed), and a narrative synthesis checked mechanically for orphan numbers. No model stage saw an answer key, a published value, or the seal; grader-side comparison happened only afterward.

2.11 Phase P5 — deployment on clean texts

The deployment phase ran the same architecture on the *original, unperturbed* texts of both anchors with no answer key in any loop — the day-to-day usage scenario. On clean texts, agreement cannot distinguish reading from recall, so this phase makes no reading-proof claim — the perturbed phases carry that burden. Its components: (i) fresh two-replicate extraction of all 21 primaries by `gemma4:12b` under the English instruments; (ii) a frozen errata-cell panel testing, at every source-confirmed erratum of Anchor 1 reachable by the sheet, whether the model's cell sides with the source or with the anchor's published (erroneous) value; (iii) deterministic recomputation of both meta-analyses and per-trial three-way tables (source quotation $\times$ the anchor authors' cell $\times$ the model's cell); and (iv) a three-configuration comparison of error detection over the same sheets — (a) the deterministic engine with no nets, (b) the engine plus deterministic sheet-versus-source nets, and (c) the model orchestrating under the conversational ten-net harness — each registered as a dated amendment before its run.

Table 1: P1 extraction under English instruments (perturbed Anchor-1 corpus; 124 eligible key cells per model). Recitation candidates are shown before/after adjudication.

Model	Cells vs key	Replicate stability	Recitation cand.	Parse failures
gemma4:12b	103/124 (83.1%)	119/124 (96.0%)	1 → 0	0
qwen3:14b	104/124 (83.9%)	123/124 (99.2%)	1 → 0	0
qwen3.5:9b	104/124 (83.9%)	103/124 (83.1%)	0	0
deepseek-r1:14b	97/124 (78.2%)	96/118 (81.4%)	1 → 0	1 replicate
llama3.1:8b	88/124 (71.0%)	96/124 (77.4%)	2 → 0	0

2.12 Comparison conventions and software

All computation used Python 3.12 with the repository's validated engine functions (risk ratio and its log-scale interval; mean difference and its interval; SD conversions; Mantel–Haenszel and DerSimonian–Laird pooling; inverse-variance pooling), each frozen and test-anchored in the pilot (the risk-ratio function reproduces a published per-study value of Anchor 1 as its standing test case); no external statistics package was used. Replicates measure stability, not statistical significance, and no formal between-model inference was attempted: with one corpus per anchor the model comparisons are descriptive, and every superlative in this article is scoped to these corpora. The campaign's compute totalled 277 model calls across P1–P4 (≈10.8 h of machine time), with every queue resume-safe and every phase committed to the public repository before it ran.

3 Results

3.1 P1 — extraction: the reading proof held, and no attributable recitation occurred

All 140 extraction calls (5 models × 14 trials × 2 replicates) completed with clean stops in 390.4 minutes, and every sheet parsed at the sheet level (one single replicate, `deepseek-r1:14b` on one trial, was unparseable; its other replicate served). Cell agreement with the corrected two-layer key ranged from 88/124 (71.0%) for `llama3.1:8b` to 104/124 (83.9%) for `qwen3:14b` and `qwen3.5:9b`, with `gemma4:12b` at 103/124 (83.1%); replicate stability ranged from 77.4% to 99.2% (Table 1). Because the numeric comparator cannot see format or granularity equivalence, cross-model ranking by this score alone is provisional; the value-level separation appears in P3.

The mechanical recitation detector raised 5 candidate cells across 140 sheets, and adjudication against the perturbed source texts resolved all 5 to incomplete-perturbation artifacts rather than recall: three models independently wrote a displaced count (113) that is arithmetically reconstructible from a surviving percentage printed beside it in the perturbed text (57.7% × n=196 = 113.1), and two models transcribed a value (4088) that survives verbatim in a prose restatement the perturbation operator missed. Zero attributable recitations remained. The same three models copied, rather than reconstructed, the equally inconsistent control-arm cell of the same table — an asymmetric preference shared across architectures, recorded as a behavior note.

3.2 P2 — arithmetic: unaided fails, the calculator repairs

Across all five models, exactly 1 of 57 unaided 95% confidence intervals was exact (arm A). With the text-protocol calculator (arm B), exact intervals rose to 41/50 (82%) among the four models that operated the tool — 11/16 for `gemma4:12b`, 14/16 for `qwen3:14b`, 8/9 for `llama3.1:8b`, 8/9 for `qwen3.5:9b` — and exact point estimates rose in step (4→12, 4→14, 5→8, 5→8). Pooled estimates remained at or near zero exact in both arms for every model

Table 2: P3, Anchor 1: DerSimonian–Laird pools from each model's own sheets (published: morbidity 0.778 [0.567, 1.068]; mortality 1.021 [0.446, 2.337]).

Model	Morbidity RR [95% CI]	Mortality RR [95% CI]
gemma4:12b	0.779 [0.569, 1.065]	1.023 [0.447, 2.344]
qwen3:14b	0.779 [0.569, 1.065]	1.021 [0.446, 2.337]
llama3.1:8b	0.771 [0.562, 1.059]	1.023 [0.447, 2.344]
qwen3.5:9b	0.779 [0.569, 1.065]	1.019 [0.445, 2.336]
deepseek-r1:14b	0.775 [0.565, 1.063]	0.980 [0.428, 2.246]

Table 3: P3, Anchor 2: the five diamonds — unperturbation-lens pooled HbA1c mean differences (published: −0.24 [−0.32, −0.16], I^2 6%).

Model	Lens MD [95% CI]	I^2	Trials in pool
gemma4:12b	−0.27 [−0.38, −0.17]	28.6%	7/7
deepseek-r1:14b	−0.47 [−1.22, 0.28]	99.1%	7/7
llama3.1:8b	−0.60 [−1.02, −0.18]	90.5%	5/7
qwen3:14b	−0.63 [−1.08, −0.17]	95.2%	6/7
qwen3.5:9b	−0.80 [−1.67, 0.07]	95.0%	6/7

in the cast, with or without the tool. `deepseek-r1:14b`'s unaided arm was unusable (5 of 6 outputs unparseable, zero gradable quantities), and its pooled-family tool arm ran without emitting a single `CALC:` line.

3.3 P3 — creation: every model reproduces the dichotomous pools; exactly one reaches the continuous diamond

From the P1 sheets, the deterministic engines reproduced Anchor 1's published pooled estimates for *all five* models: pooled morbidity 0.771–0.779 against the published 0.778 [0.567, 1.068], and pooled mortality 0.980–1.023 against the published 1.021 [0.446, 2.337], every value within ±0.01 of the published except one mortality estimate at Δ0.04 (Table 2). The postoperative-ileus pool is non-comparable by construction (anchor erratum #16: one published row is a different outcome) and was reported, never pooled against.

From the fresh Anchor-2 extractions (70/70 calls, 155.1 minutes), the same engine and sealed lens produced five diamonds (Table 3). `gemma4:12b`'s lens landed at −0.27 [−0.38, −0.17] (I^2 28.6%), beside the published −0.24 [−0.32, −0.16] — and identical to the hundredth to the same model's result in the archived development pilot. The other four landed between −0.47 and −0.80 with saturated heterogeneity (I^2 90.5–99.1%), two of them with starved sheets (5/7 and 6/7 trials in the pool).

3.4 P4 — orchestration under the English harness

`gemma4:12b` closed all 7 per-study call sequences under the English ten-net harness in 8.7 minutes. Two content warnings fired — both on one trial, both from the declared-derivation net — and both were resolved by the confirm-or-correct mechanism; no other net fired and no format failure occurred. The model then emitted the pooling call over its own seven sextets and closed digit-consistently with the executed result (−0.50 [−0.78, −0.22]); the mechanical truth over the same sheets was −0.52 [−0.82, −0.22] (ΔMD 0.02, from two named route divergences). The product layer flagged zero incoherent intervals, and the synthesis contained zero orphan numbers. The grader-side lens over the same sheets was −0.27 [−0.38, −0.17].

Table 4: P5: three error-detection configurations against the same single-cell defect, same sheets, no answer key in any loop.

Configuration	Pool delivered	The poisoned cell
(a) deterministic engine, no nets	-0.34 [-0.51, -0.18]	silent
(b) engine + sheet-vs-source nets	-0.34 (unchanged; warn-only)	flagged (exp. arm, with quotes)
(c) model + conversational ten-net harness	-0.32 [-0.46, -0.19]	not flagged; 4 live slips caught

3.5 P5 — deployment on clean texts: the errata are visible, and the one miss is one cell

On the original, unperturbed texts, replicate agreement was 96.8% (Anchor 1) and 95.4% (Anchor 2), and all 42 sheets parsed. At every cell of the frozen errata panel — 7 of 7, including all four direction-critical swaps — the model's extraction sided with the *source* against the anchor's published value; on the trial whose published ileus row reproduces its pulmonary-complication counts (erratum #16), the model wrote *not reported* in both replicates rather than reciting the published numbers. The deterministic recomputation reproduced Anchor 1's pooled morbidity (0.779 [0.569, 1.065]) and mortality (1.019 [0.445, 2.336]) within tolerance; the Anchor-2 pool missed its tolerance at -0.34 [-0.51, -0.18] against the published -0.24 [-0.32, -0.16], and the residue decomposed entirely to one cell — the model transcribed a printed "-1.6 ± 0.3" from one trial's results prose and declared the $\pm$ a standard deviation, where the value equals the half-width of the confidence interval printed in the same trial's table. Correcting that single cell by the key's own conversion rule yields -0.24 [-0.32, -0.16] (I^2 3.7%), the published diamond to the hundredth; the cell was identical in both replicates, and zero invented values occurred in the phase's 124 graded cells.

Over the same sheets, the three detection configurations behaved differently against that poisoned cell (Table 4): the bare engine delivered the distorted pool silently; the sheet-versus-source net flagged the experimental arm mechanically, with both source quotations, after a first rule revision (its initial form stayed correctly silent because the trial's own prose prints the $\pm$ form — the flag rule was extended, as a dated amendment, to fire when a printed $\pm$ equals the printed interval's half-width; the control arm's $\pm$ matches nothing within the pre-registered tolerance and was not flagged); and the conversational ten-net harness raised zero flags on the poisoned cell — whose downstream calls were well-formed, correctly source-declared and internally coherent — while catching four live orchestration slips (one arity, three derivation declarations) in the same run. A DerSimonian–Laird weight-share flag (>40%) was blind to the distortion (the poisoned trial's share was 19.3%; the distortion's product signature was heterogeneity inflation, I^2 6%→79%).

3.6 Instruction-language robustness

Per phase, the campaign's English-instrument results sat within replicate-level variance of the archived Portuguese-instrument pilot (Table 5). The largest reading-stage difference was +3 cells of 124 on the one directly comparable model, with replicate stability identical to the decimal; the arithmetic signature (unaided failure, calculator repair, pool ceiling, per-model deviant classes) matched point for point; the creation-stage flagship value was identical to the hundredth; and the orchestration-stage delta sat inside the measured route-variability band with the same warning classes on the same trial.

Table 5: Instruction-language robustness: the English-instrument campaign vs the archived Portuguese-instrument development pilot, per phase (directly comparable quantities only).

Phase	Campaign (English)	Pilot (Portuguese)
Read (gemma4:12b, cells / stability)	83.1% / 96.0%	80.6% / 96.0%
Calculate (unaided exact CIs; tool arm)	1/57; 82% exact	0/30; same signature
Create (gemma4:12b lens, Anchor 2)	−0.27 [−0.38, −0.17]	−0.27
Orchestrate (pool vs truth; behavior)	Δ0.02; confirm-or-correct	Δ0.06; same classes

4 Discussion

This study measured, in one continuous pre-registered chain, what small local models running on an integrated GPU can contribute to meta-analysis — and where the risk concentrates. The models read clinical trials reliably enough to rebuild two published meta-analyses; they cannot do the interval arithmetic unaided but do it almost perfectly through a trivial text-protocol tool; a deterministic architecture turns their sheets into pooled estimates that match the published ones; and exactly one model in the class produces a continuous-outcome diamond beside the published value. Under the perturbation control, none of this is attributable to memory of the published review.

Because the instruments were developed in Portuguese and the campaign ran in English over otherwise identical materials, the study carries its own instruction-language robustness check. A reasonable referee could ask whether either language handicapped — or peculiarly helped — models trained predominantly on English; the comparison answers by measurement rather than argument. The one directly comparable reading score moved by three cells in 124, inside the format-noise band, with replicate stability identical to the decimal; the arithmetic signature reproduced point for point; and the flagship creation-stage value was −0.27 under both languages, identical to the hundredth. This is consistent with multilingual instruction-following being a solved sub-problem at these model sizes for structured extraction tasks, and it removes instruction language as a confounder for every conclusion reported here.

The five-model comparison exposes a clean division of difficulty. Every model in the cast — including the weakest reader — delivered sheets from which the engines reproduced the published *dichotomous* pools: counting events and denominators is within reach of the whole class. The *continuous* chain is what separates models: means, dispersions and, above all, correctly identifying what a printed dispersion *is*. The five diamonds spread from −0.27 to −0.80 entirely through reading-class differences, never arithmetic — the same engine computed all five — supporting the study's central design thesis: once everything deterministic is code, reading is the whole game, and model choice is a reading choice. The pooled-arithmetic ceiling points the same way: even with the calculator, assembling the list-valued pooling calls stayed beyond every model in this size class (in the archived pilot, only a 27-billion-parameter model ever mastered that call) — which is precisely why the architecture moves pooling into code rather than teaching it. For practitioners this suggests a concrete triage: dichotomous evidence tables may be delegated broadly, while continuous outcomes concentrate all the model risk and deserve the strongest reader plus the strongest checks.

The deployment phase turns the benchmark toward real use, where no answer key exists. Its clean-text results support the day-to-day scenario in three ways. First, the model's extractions sided with the primary sources at all seven source-confirmed errata of the published meta-analysis — including refusing, in both replicates, to supply the ileus counts that the anchor publishes but its source never reports — so a local reader plus a three-way table (source quotation, human cell, model cell) makes published extraction errors *visible* to a reader rather than requiring trust in either extractor. This is consistent with reports that language-model assistance can match human extraction quality in live reviews [Lee et al., 2026], and it adds the complementary observation that the disagreements are informative in both directions: the same source-verification process also corrected the benchmark's own key twice during development, and the errata file carries the adjudicators' own published errors. Second, the phase's one failed tolerance decomposed to a single cell with an instructive anatomy: the primary itself prints the same quantity in two layers — a table interval and a prose "±" that equals the interval's half-width — and the model transcribed the prose layer and mislabeled the ± as a standard deviation, identically in both replicates. The defect is seeded by the source's own ambiguous notation, and it defines this architecture's characteristic residual failure class: a model that misdeclares its sheet escapes every check that trusts the sheet. Third, the three-configuration comparison shows that detection layers are complements, not substitutes: conversational nets catch conversation-layer defects (four live orchestration slips in a single run) but are structurally blind to a well-formed poisoned cell, which only a sheet-versus-source net saw — and a weight-share product flag was blind too, because random-effects weighting re-equalizes shares while the distortion surfaces as heterogeneity inflation. No warn-only configuration made the wrong number right — by doctrine it must not — so the safest key-free architecture this record supports is a deterministic engine, sheet-versus-source detection nets, a conversational harness for orchestration slips, and a human dispatched to the primary's table whenever any flag fires.

These results sit within a fast-moving literature. Feasibility studies report promising but uneven extraction accuracy for cloud models [Schmidt et al., 2024]; screening assistance is approaching routine [Xie et al., 2026]; and end-to-end platforms report complete reviews in hours [Bakker et al., 2026, McLaughlin et al., 2026]. In contrast to that line, everything measured here runs locally, at zero marginal cost, on hardware compatible with the privacy constraints of clinical text [Wiest et al., 2024, Spaanderman et al., 2025] — and under evidence-synthesis organizations' stated conditions of demonstrated rigor and transparent reporting [Flemyng et al., 2025], which the perturbation proof, the public seals and the adjudication rite are designed to operationalize. The reading-versus-recall control matters more, not less, as anchors and primaries enter training corpora [Xu et al., 2024]: in 140 perturbed sheets the campaign found zero attributable recitations, while both mechanical candidates it did raise traced to survivable gaps of the perturbation operator itself — now demonstrated across multiple architectures and queued as instrument fixes.

Several limitations bound these claims. Two anchors from a single journal family and 21 primaries are the whole evidence base; every superlative is scoped to these corpora, and no formal between-model inference is attempted. One quantization per model and one hardware profile were tested. The mechanical cell comparator is a declared approximation of the full adjudicated ruler, so absolute cell scores understate agreement and cross-model gaps partly reflect format habits. The perturbation operator's residual gaps (derivable totals; prose restatements) are documented and exploitable, though they produce reconstruction, not recall. The English harness build reproduces the frozen net logic but was exercised in one formal run of one model. The adjudicator is the same human–AI pair that built the harness, mitigated by mandatory quotation and public self-errata rather than independence, and the known biases of model judges [Zheng et al., 2023] are constrained, not eliminated, by that rite. Finally, the deployment phase's clean texts cannot separate reading from recall by design; its claims rest on the perturbed phases' proof.

In conclusion, a consumer machine running one well-chosen 12-billion-parameter local model, a frozen deterministic engine and warn-only detection nets can rebuild published meta-analyses to the published digits, expose documented errors in them with source quotations, and fail in ways that are named, single-celled and flaggable — and none of this depends on the instruction language. The designed next step is the generalization gate: the same frozen architecture against a new anchor in a new clinical domain, where transfer, not replication, is the question.

Data and code availability

All protocols (pre-registered, with dated amendments committed before each run), the frozen English instrument library with its Portuguese-correspondence tables, perturbation seals with SHA-256 fingerprints, the two-layer answer keys and the anchor errata file, the deterministic engines and graders, the English harness build, all 277 raw model outputs and run logs, per-phase evaluation records, adjudications with source quotations, and the exact quantized builds (pinned by Ollama digest) are public at `https://github.com/gbbarra/llm-evidence-extraction-benchmar` and archived at Zenodo [Barra, 2026]. Closed-stratum primary articles are not redistributed; scripts rebuild that corpus stratum from legally obtained copies.

Declarations

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